

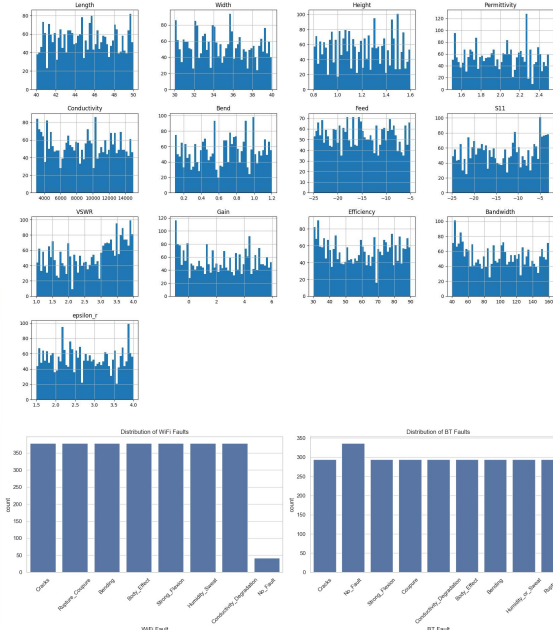
Antenna Performance and Fault Detection

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Introduction

The goal of this project is to build a classifier model with capable of fault detection in antennas. Based on a series of 13 different features, the model must be able to predict whether or not the Bluetooth, wifi or both are failing. This problem is thus a multioutput classification problem with supervised learning. The main chosen method is the Random Forest Classifier but the Extra Trees Classifier and Linear SVC models are also tested.

Materials and methods



Results

Standard RFC

Accuracy Score: 0.8829
Jaccard Score: 0.7928
F1 Score: 0.8829

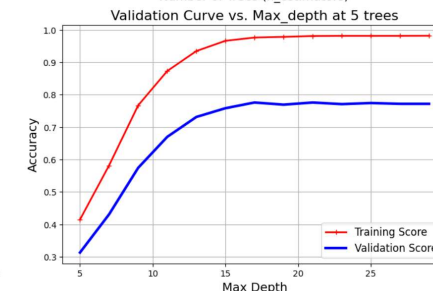
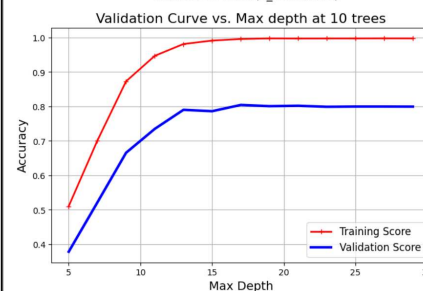
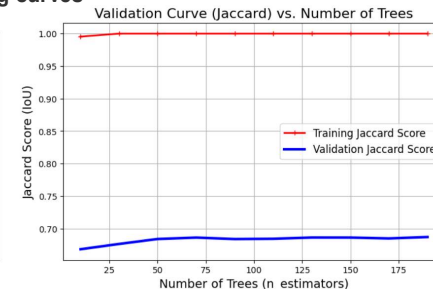
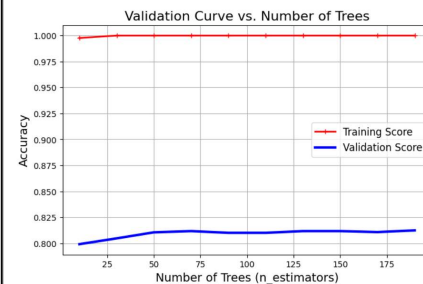
Standard ETC

Accuracy Score: 0.8810
Jaccard Score: 0.7886
F1 Score: 0.8806

Standard Linear SVC

Accuracy Score: 0.2203
Jaccard Score: 0.1149
F1 Score: 0.2022

Learning curves



Hyperparameters:

```
tuning_parameters={
    "n_estimators": [200, 400, 600, 800, 1000],
    "max_depth": [None, 10, 20, 40, 60, 80, 100],
    "min_samples_split": [2, 5, 10, 20],
    "min_samples_leaf": [1, 2, 4, 8],
    "max_features": ["sqrt", "log2", 0.3, 0.5, None],
    "class_weight": [None, "balanced", "balanced_subsample"],
    "criterion": ["gini", "entropy"]
}
```

Loss/impurity functions

$$\text{Gini} = \sum_{k=1}^K p_k(1 - p_k)$$

$$\text{Entropy} = - \sum_{k=1}^K p_k \log(p_k)$$

Grid Search Optimization

model=RandomForestClassifier(bootstrap=True,class_weight='balanced',criterion='gini',max_depth=60,max_features='sqrt',min_samples_leaf=2,min_samples_split=5,n_estimators=400)

Flat string accuracy: 0.88755
Exact pair accuracy: 0.86059
Jaccard Score: 0.7968
F1 Score: 0.8860

Conclusions

This project demonstrates that the choice of machine learning model significantly impacts predictive performance. To select the optimal model, an engineer must possess a deep understanding of the data, prioritize exploratory data analysis, and perform effective preprocessing, such as the inclusion of the 'volume' feature and train-test splitting.

Among the three evaluated models (LinearSVC, ExtraTreesClassifier, and RandomForestClassifier), the RandomForestClassifier initially emerged as the superior choice. Subsequent hyperparameter tuning via GridSearch further refined the results, leading to final scores of: Accuracy: 0.8876, Jaccard: 0.7968, and F1: 0.8860. These results confirm the model's capability to accurately predict antenna fault causes based on specific technical specifications.

Further information

Feature augmentation was used to improved the accuracy by adding the feature volume, since length, width and height was known features.

Literature cited

- Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (3rd ed.). O'Reilly Media. p. 116. One-hot encoding
- Scikit-learn.org -1.10.7 Mathematical formulation (<https://scikit-learn.org/stable/modules/tree.html#tree-mathematical-formulation>) - Loss function
- Scikit-learn.org - RandomForestClassifier (<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>)
- !20-searching is used, where searchReport prints the information about the best model, Classification_report tests the best model and FullReport prints both.